Safety Ride Helmet

Implementation of an IoT-enabled smart helmet with real-time drowsiness detection, speed monitoring, and voice-guided safety systems.

- Team:
- K.M.P.N. Karunaratne
- R.M.P.B. Karunasekara
- H.M.T.H. Lenadora
- G.W.T. Senesh.

Problem Statement

Human Error

Many motorcycle accidents stem from preventable errors like over-speeding or microsleeps.

Detection Gap

Standard gear cannot reliably detect drowsiness or microsleep episodes.

Speed Risk

Unintentional speeding increases accident severity.

Delayed Rescue

Solo crashes often lose critical time because the location can't be communicated.

Project Objectives

Design and prototype a Smart Safety Helmet that detects drowsiness, monitors speed, and provides voice-guided safety with emergency alerts.

Drowsiness Detection

IR sensor-based monitoring of eye-blink patterns to identify sleepiness.

Speed Monitoring

GPS-derived real-time velocity triggers voice warnings when thresholds are exceeded.

Crash Alerts

Tilt/impact detection locks GPS coordinates and sends SMS to emergency contacts.

Hardware Architecture

Core components that make the helmet autonomous and communicative.

Microcontroller

Arduino / ESP32 for control and processing.

IR Sensor

Monitors eye closure duration for fatigue.

Tilt / Accelerometer

Detects abnormal orientation or impact.

NEO-6M GPS

Provides coordinates and real-time speed.

SIM800L GSM

Sends emergency SMS with location.

DFPlayer Mini + Speaker

Stores and plays voice prompts and alerts.

Software Logic

01

Continuous Monitoring

Loop analyzes eye-closure duration to trigger wake-up voice alerts.

02

Speed Analysis

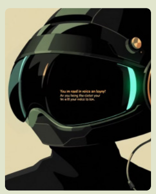
Real-time GPS speed checks issue verbal warnings when limits are exceeded.

03

Emergency Routine

Detects crash via tilt sensor, locks GPS, and dispatches SMS to contacts.

Functional Features

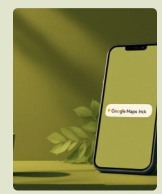


Voice Coaching

Reminders for lights and speed compliance through clear voice prompts.

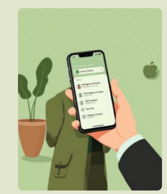
Wake-up Alerts

Immediate wake-up prompts to prevent accidents from exhaustion.



Automated SMS

Emergency contacts receive exact GPS coordinates and a Google Maps link.



User Control

Users can choose emergency contacts via mobile App

These features combine preventive guidance with rapid, automated emergency response to reduce injury and fatality risk.

Feasibility & Conclusion

Standalone Design

No integration with motorcycle wiring required; fits any vehicle.

Rechargeable Power

Sensors are powered by an internal rechargeable battery housed in the helmet.

Active Safety

Combines voice guidance and reactive emergency alerts to lower fatality probability.

Shift in Protection

Represents a move from passive protection to intelligent rider assistance.

Future Improvements

Lighter Design & Better Battery

Upgrading to a smaller, more efficient computer chip to reduce the overall weight of the helmet and make the battery last longer.

Smarter Sleep Detection

Improving the drowsiness feature with smart software (AI) that learns the rider's unique blinking and fatigue patterns for much more accurate alerts.

Secure Ride History

Creating a secure online system to save ride statistics, so users can look back at their habits over time, like how often they speed.

Smartphone App

Building a mobile app that lets riders easily connect to the helmet to change their speed limits, update their emergency contacts, and view their past rides right from their phone.

Thank You!